Answers and marking guidance

Award one mark per correct, developed point unless stated. Other valid answers should be credited.

2 Calculate Vout.

- Vout = −(Rf ÷ Rin) × Vin = −10 × 0.35
- Vout = −3.5 V

3 Quick check

- a) Voltage follower
- b) 10.2 V
- c) −2 V
- d) V+ > V−

4 Key terms

- 1 – F
- 2 – B
- 3 – D
- 4 – A
- 5 – E
- 6 – C

5 Calculate Rf.

- Gain = 1 + Rf ÷ Rin, so 11 = 1 + Rf ÷ 1000
- Rf = 10 kΩ

6 Calculate Vout.

- Vout = −Rf × (V1/R1 + V2/R2) = −22k × (0.2/10k + 0.5/10k)
- Vout = −1.54 V

7 Calculate Vout.

- Vout = (V2 − V1) × Rf ÷ R1 = 0.6 × 3.3
- Vout = 1.98 V (2.0 V)